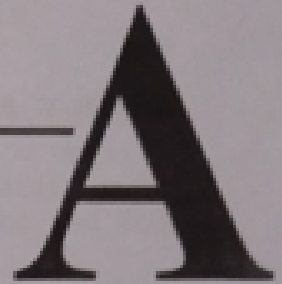


No. of Pages	4
No. of Questions	4
Total Marks	25
Time: 70 minutes	

Department of Computer Science and Engineering
Midterm Examination Spring 2026
CSE 110: Programming Language I



No washroom breaks

- ✓ Write your **Set Number** on top of the answer script in **LARGE FONT**.
- ✓ Answer all questions. Use **back part** of the answer script for rough work.
- ✓ Answer Question 1 & 2 at the **beginning part** of the answer script.
- ✓ Figure in bracket [] next to each question indicates marks for that question.
- ✓ At the end of exam, put **question paper** inside answer script and **return both**.
- ✓ Understanding the question is part of the exam, **please do not ask questions**

Section: _____ ID: _____ Name in CAPITAL: _____

Question 1 [CO1] [10 Points]
[Answer on the answer-script]

A monitoring system receives odd-length positive integer data packets (where length > 2). A packet is non-corrupted if the sum of the left digits (digits before the middle digit) is greater than the sum of the right digits (digits after the middle digit). The middle digit is ignored. The system keeps accepting packets until 5 non-corrupted packets are detected. After the 5th non-corrupted packet, it stops and computes the percentage of corrupted packets.

- If less than 25% of the packets are corrupted, the system is considered functional.
- Otherwise, the system indicates that maintenance is required.

Write a Java program that takes multiple inputs and identifies the percentage of corrupted data. Assume the user will always input an odd-length positive data packet (where length>2).

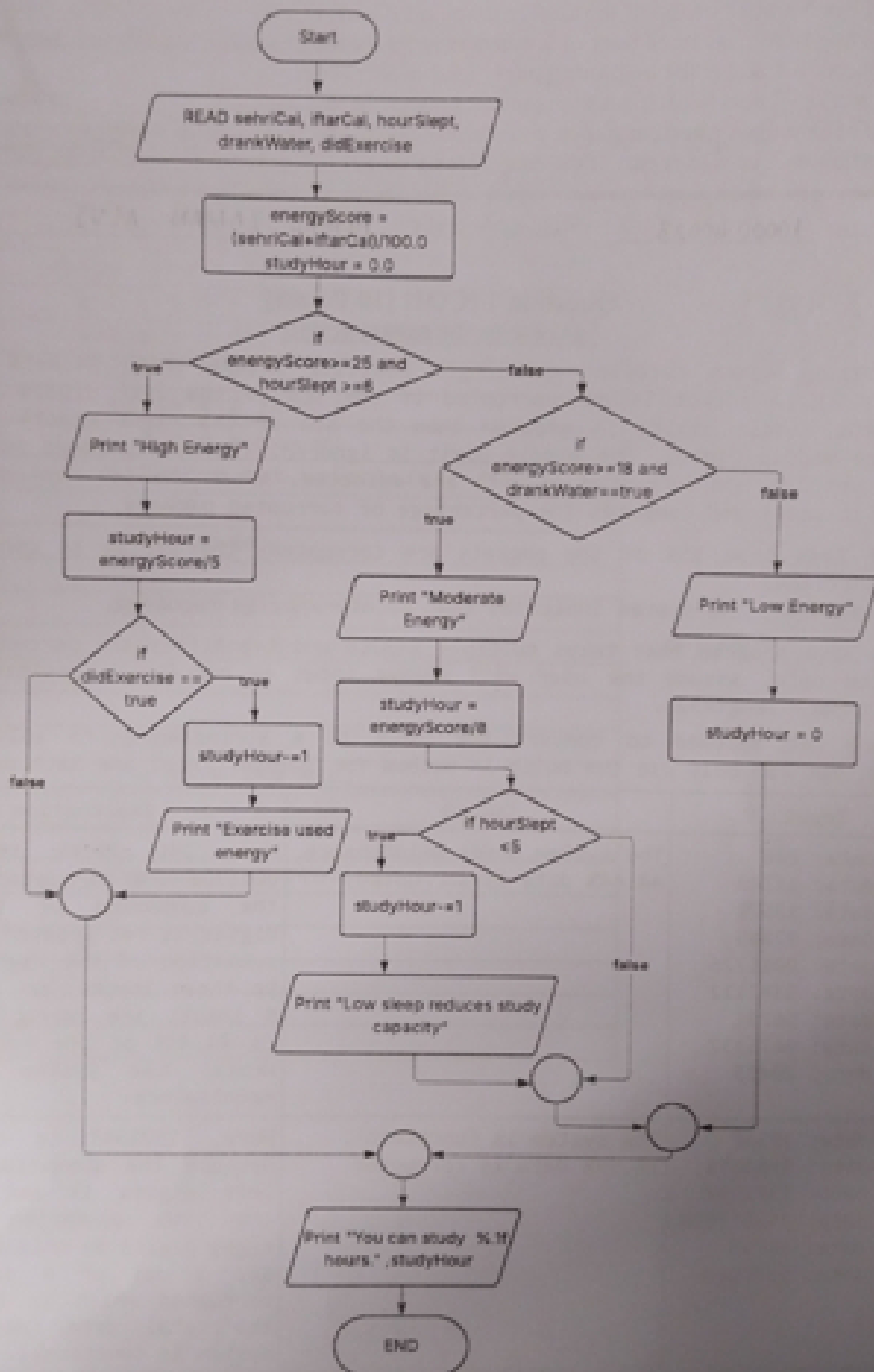
[You are not allowed to convert the input to a String/Array to solve this problem. You can only use the built-in method for integer input and Math.pow()]

Input	Output	Explanation
Enter data: 246 Enter data: 12345 Enter data: 13579 Enter data: 92345 Enter data: 9811726 Enter data: 3342112 Enter data: 56781 Enter data: 6675332 Enter data: 29435	The system needs maintenance. 44.44% data is corrupted.	Here, 246, 12345, 13579, and 9811726 are corrupted because the summation of the left digits is not greater than the summation of the right digits in these inputs. So, 4 out of 9 inputs are corrupted which is 44.44% of the total data. Hence, the system requires maintenance.
Enter data: 19340 Enter data: 9983658 Enter data: 3321547 Enter data: 246813570 Enter data: 98765 Enter data: 5674123	The system is functional. 16.67% data is corrupted.	Here, 3321547 is corrupted because the summation of the left digits is not greater than the summation of the right digits in this input. So, 1 out of 6 inputs is corrupted which is 16.67% of the total data. Hence, the system is functional.

Question 2 [CO3] [6 Points]

[Answer on the answer-script]

Implement a Java program from the given flowchart -



Question 3 [CO1] [5 Points]

[Answer on the question paper]

A cricket fan wants to track a short match. The user will enter the number of overs, and for each over, the runs scored. If an over has 15 or more runs, the program prints "Great over!". If 0 runs are scored in an over (maiden over), it counts as a wicket. At the end, the program prints the total runs and total wickets lost.

There are 6 errors in this code. Figure out the errors and write the corrected lines in the box below. One correction has been done for you. Do not entirely omit a code part; just correct the given code.

1	import java.util.scanner;
2	public class CricketMatch {
3	public static void main(String[] args){
4	Scanner sc = new Scanner(System.in);
5	int runs = 0;
6	int wickets = 0;
7	System.out.print("Enter number of overs: ");
8	int overs = sc.nextInt();
9	for(int i = 1; i <= overs; i++){
10	System.out.print("Runs in over " + i + ": ");
11	int r = input.nextInt();
12	runs += r;
13	if(r > 15){
14	System.out.println("Great over!");
15	}
16	if(r == 0){
17	wickets--;
18	}
19	}
20	System.out.println("Total runs: " + runs);
21	System.out.println("Wickets lost: " + wickets);
22	}
23	}

Write your corrections in the table given below. For better understanding, one error correction is shown.

Line Number	Fix
Line 1	import java.util.Scanner;
Line 9	for (int i = 1; i <= overs; i++) {
Line 11	int r = sc.nextInt();
Line 13	if (r >= 15) {
Line 14	System.out.println("Great over!");
Line 17	wickets++;

Question 4 [CO1] [4 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

```
1 public class A{
2     public static void main(String args[]){
3         int p = 0, q = 7, m = 10;
4         boolean flag = true;
5         while (p++ < 2){
6             q = p % 9 + 12; —
7             while(q > 4){
8                 int c = 7;
9                 if (p % 2 != 0){
10                     c = q + m - p;
11                     System.out.println(c); —
12                     q-=8;
13                     continue;
14                 }
15                 else if(!flag){
16                     c += p + q + 16;
17                     break;
18                 }
19                 else{
20                     m += p % q + c;
21                     c = m + c % 10;
22                 }
23                 System.out.println(c + m); —
24                 q-=7;
25             }
26         }
27     }
28 }
```

Output
22 ✓
25 ✓
60 ✓
74 ✓

26...16

10 + 16 + 10

No. of Pages	4
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B

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- ✓ At the end of exam, put **question paper** inside answer script and **return both**.
- ✓ Understanding the question is part of the exam, please do not ask questions

No washroom breaks

Section: [REDACTED] ID: [REDACTED] Name in CAPITAL: [REDACTED]

Question 1 [CO1] [10 Points]

[Answer on the answer-script]

A monitoring system receives odd-length positive integer data packets (where length > 2). A packet is non-corrupted if the sum of the left digits (digits before the middle digit) is greater than the sum of the right digits (digits after the middle digit). The middle digit is ignored. The system keeps accepting packets until 5 non-corrupted packets are detected. After the 5th non-corrupted packet, it stops and computes the percentage of corrupted packets.

- If less than 25% of the packets are corrupted, the system is considered functional.
- Otherwise, the system indicates that maintenance is required.

Write a Java program that takes multiple inputs and identifies the percentage of corrupted data. Assume the user will always input an odd-length positive data packet (where length > 2).

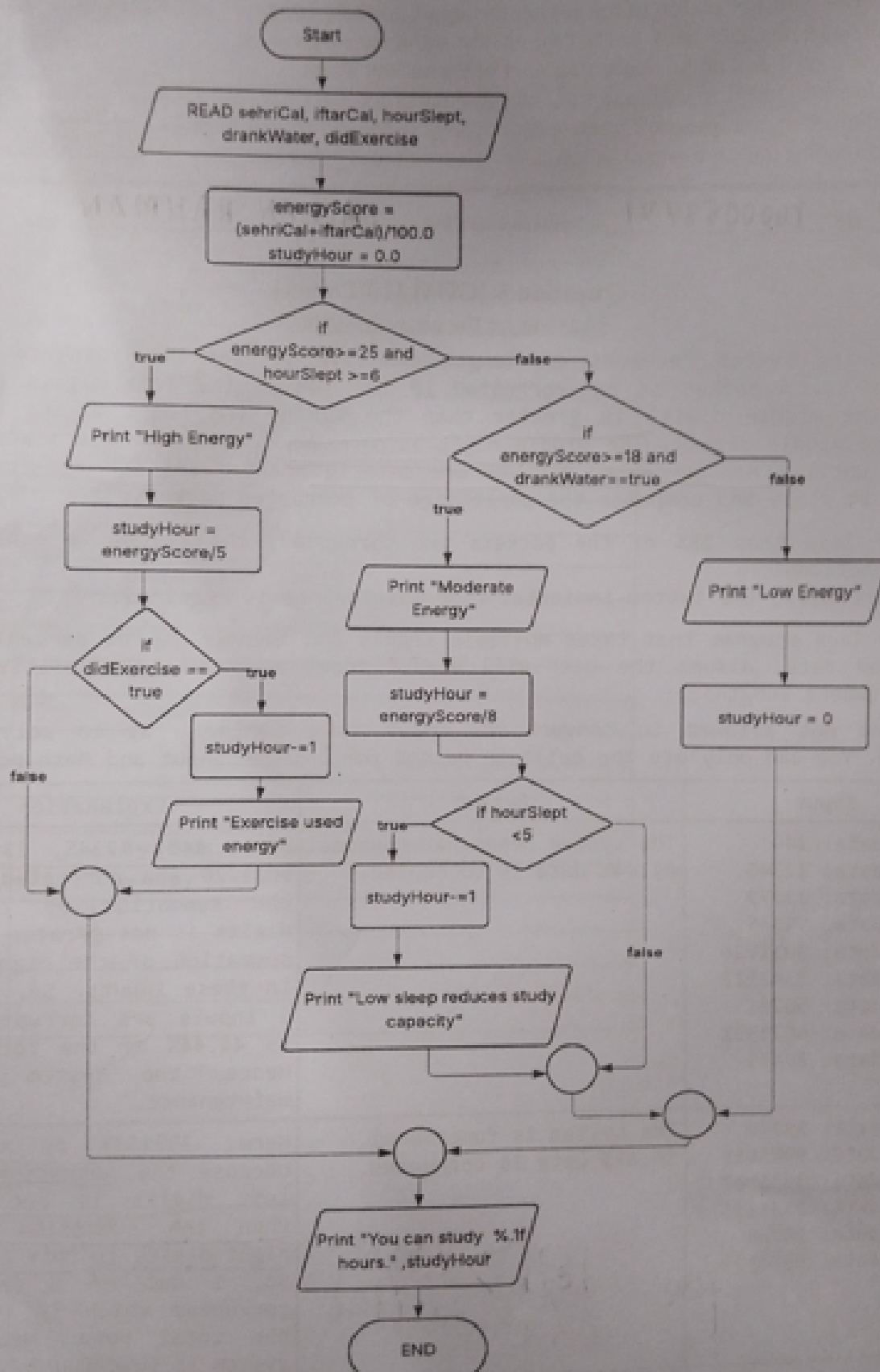
[You are not allowed to convert the input to a String/Array to solve this problem. You can only use the built-in method for integer input and Math.pow()]

Input	Output	Explanation
Enter data: 246 Enter data: 12345 Enter data: 13579 Enter data: 92345 Enter data: 9011726 Enter data: 3342112 Enter data: 56781 Enter data: 6675332 Enter data: 29435	The system needs maintenance. 44.44% data is corrupted.	Here, 246, 12345, 13579, and 9011726 are corrupted because the summation of the left digits is not greater than the summation of the right digits in these inputs. So, 4 out of 9 inputs are corrupted which is 44.44% of the total data. Hence, the system requires maintenance.
Enter data: 19340 Enter data: 9983658 Enter data: 3321547 Enter data: 246813570 Enter data: 98765 Enter data: 5674123	The system is functional. 16.67% data is corrupted. <i>Left > Right ↳ corrupted</i>	Here, <u>3321547</u> is corrupted because the <u>summation of the left digits is not greater than the summation of the right digits in this input.</u> So, 1 out of 6 inputs is corrupted which is 16.67% of the total data. Hence, the system is functional.

Question 2 [CO3] [6 Points]

[Answer on the answer-script]

Implement a Java program from the given flowchart -



A cricket fan wants to track a short batting practice session. The user will first enter the number of balls faced, and then input the runs scored from each ball one by one. If a ball results in 4 or more runs, the program should print "Boundary!". If 0 runs are scored, it will be counted as a dot ball. After all the balls are entered, the program should display the total runs scored and the total number of dot balls.

There are 6 errors in this code. Figure out the errors and write the corrected lines in the box below. One correction has been done for you. Do not entirely omit a code part; just correct the given code.

1	import java.util.scanner;
2	public class BattingPractice {
3	public static void main(String[] args){
4	Scanner sc = new Scanner(System.in);
5	int runs = 0;
6	int dots = 0;
7	System.out.print("Enter number of balls: ");
8	int balls = input.nextInt();
9	for(int i = 1; i < balls; i++){
10	System.out.print("Runs in ball " + i + ": ");
11	int r = sc.nextInt();
12	runs += r;
13	if(r > 4){
14	System.out.println(Boundary!);
15	}
16	if(r == 0){
17	dots--;
18	}
19	}
20	System.out.println("Total runs: " + runs);
21	System.out.println("Dot balls: " + dots);
22	}
23	}

Write your corrections in the table given below. For better understanding, one error correction is shown.

Line Number	Fix
Line 1	import java.util.Scanner;
Line 8	int balls = sc.nextInt(); ✓
Line 14	System.out.println("Boundary!"); ✓
Line 17	System.out.print dots++; ✓
Line 13	if (r == 4) ✓
Line 9	for(int i = 1; i <= balls ; i++) { ✓

Question 4 [CO1] [4 Points][Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

1	public class B{
2	public static void main(String args[]){
3	int p = 0, q = 7, m = 0;
4	boolean flag = true;
5	while (p++ < 2){
6	q = p % 5 + 12;
7	while(q > 4){
8	int c = 3;
9	if (p % 2 != 0){
10	c = q + m - p;
11	System.out.println(c);
12	q-=8;
13	continue;
14	}
15	else if(!flag){
16	c += p + q + 10;
17	break;
18	}
19	else{
20	m += p % q + c;
21	c = m + c % 5;
22	}
23	System.out.println(c + m);
24	q-=7;
25	}
26	}
27	}
28	}

Output
22
24
45
63